



ACCT 5034 Analysis and Visualisation of Financial Data
Assessment THREE: Tableau Story & Business Analyst Report– Marking
Guide (50%)
Semester 1, 2026

Background

This assessment integrates advanced Tableau skills with critical business analysis. You will work with a panel dataset of Asia-Pacific listed companies covering nine economies — Australia, Bangladesh, China, India, Indonesia, Republic of Korea, Malaysia, New Zealand and Singapore — over the period 2010–2024. The dataset, extracted from London Stock Exchange Group (LSEG) Workspace and Compustat Global, includes firm-level financial indicators, Environmental, Social and Governance (ESG) scores and pillar sub-scores, ESG controversies, and corporate governance variables.

ESG and governance are now central to investment analysis, corporate decision-making and financial regulation across the Asia-Pacific. Mandatory climate-related disclosure regimes are now in force or being implemented in Australia (AASB S1/S2 from January 2025), New Zealand (NZ CS standards under XRB, mandatory from 2023), Singapore (SGX climate-related disclosure rules), Hong Kong/China (HKEX and CSRC sustainability disclosure guidelines), Japan and Korea (KSSB), and India (BRSR Core for the top 1,000 listed entities). Regulators including ASIC, AASB, the NZ Financial Markets Authority (FMA), Monetary Authority of Singapore (MAS), Securities and Exchange Board of India (SEBI), Reserve Bank of Australia (RBA) and the China Securities Regulatory Commission (CSRC), Hong Kong Exchanges and Clearing (HKEX), and the Korea Sustainability Standards Board (KSSB) are strengthening disclosure frameworks and anti-greenwashing enforcement, broadly aligning with the IFRS Foundation's ISSB standards.

Your task is to:

- **Develop a Tableau Story** with exactly **three dashboards** that present a coherent, stepwise narrative.
- Produce a 1,000-word **Business Analyst Report** that interprets your Tableau outputs, integrates academic and practitioner sources, and provides critical recommendations.

This task simulates professional work undertaken by business analysts, ESG/sustainability consultants and investment researchers.

Dataset Snapshot

The raw panel contains **237,506 firm-year** observations across **16,322 unique firms** (2001–2024). ESG scores are only available for a subset of larger, listed firms (**≈17,368 firm-year** observations). For this assessment you must filter the data in Tableau to firm-years that have a non-null ESG Score (ESGS). This makes the workbook performant and ensures every visual is built on a comparable, ESG-disclosing sub-sample. Country-level coverage of the full panel is shown below; ESG coverage is concentrated in larger firms within each market.

Country of Exchange	Firm-years	% of panel	Cumulative %
Australia	23,279	9.80%	9.80%
Bangladesh	3,134	1.32%	11.12%
China	75,336	31.72%	42.84%
India	69,594	29.30%	72.14%
Indonesia	10,984	4.62%	76.77%
Korea, Rep. (S. Korea)	24,832	10.46%	87.22%
Malaysia	18,441	7.76%	94.99%
New Zealand	1,867	0.79%	95.77%
Singapore	10,039	4.23%	100.00%
Total	237,506	100.00%	

Key Variables (selected)

Code	Variable	Type / Scale	Notes
ESGS	ESG Score (overall)	0–100	Use as primary ESG measure and as a data filter (not null)
EPS	Environmental Pillar Score	0–100	Component of ESGS
SPS	Social Pillar Score	0–100	Component of ESGS
GPS	Governance Pillar Score	0–100	Component of ESGS
CONT	ESG Controversies Score	0–100 (higher = fewer/less severe)	LSEG/Refinitiv inverse scale; <100 indicates a controversy occurred
BS	Board Size	Count (1–28)	Number of directors

Code	Variable	Type / Scale	Notes
FOB	Female on Board	% (0–100)	Already a percentage — do NOT divide by BS
IBM	Independent Board Members	% (0–100)	Already a percentage of independent directors
CEOD	CEO Duality	Binary 0/1	1 = CEO is also Chair
MCAP	Market Capitalisation	USD	End-of-period; use for size and Tobin's Q
at	Total Assets	USD millions	Compustat Global
lt	Total Liabilities	USD millions	Compustat Global
ceq	Common/Ordinary Equity	USD millions	Used for MTB
sale	Sales/Turnover (Net)	USD millions	
ebit	Earnings Before Interest & Tax	USD millions	Use for ROA = ebit / at
ebitda	EBITDA	USD millions	Alternative profitability measure
GICSIndustryName	GICS Industry Name (53 industries)	Text	Use for industry breakdowns
GICSIndustryGroupName	GICS Industry Name (24 groups)	text	Use for industry breakdowns
CountryofExchange	Country of Exchange	Text	Use as primary country filter
year	Fiscal year	2001–2024	Filter to 2010–2024 for ESG-rich window

Assessment Submission Instructions

All submissions must be made via Blackboard. Two separate files are required:

1. Tableau Packaged Workbook (.twbx)

- Save your Tableau Story as a Packaged Workbook.
- Upload the .twbx file directly to Blackboard.
- This file will be used to verify the technical accuracy of your dashboards (filters, parameters, forecasts, calculations).
- File name format: StudentID_LastName_Viz.twbx

2. Business Analyst Report (.docx)

- You must submit a written report of approximately 1,000 words ($\pm 10\%$) in Word format.

The report must:

- Follow the required structure (Executive Summary, Introduction, Findings & Analysis, Recommendations, Conclusion).
- All sources must be referenced using an appropriate academic style (e.g., APA).
- Contain a working **Tableau Public link** to your Story (compulsory).¹
- The link must be included prominently in the cover page of the report.
- A missing or non-functional link will be treated as a non-submission of the Tableau Story
- **File name format:** StudentID_LastName_Report.docx

The **cover page** of the report must include Student ID, Full Name, Unit Code, Assessment Title and **Working Tableau Public Link**

Important Notes

- Both submissions are compulsory. Failure to submit either the Tableau Packaged Workbook or the Business Analyst Report will result in a mark of zero.
- Students are expected to consult Appendix A (Data Preparation in Tableau) and Appendix B (Suggested Sources for Support).

¹ When publishing your Tableau Story, you must use **Tableau Public**. After publishing, click the **Share** icon and copy the **URL link** (not the embed code). Before submission, test this link in an incognito browser window to confirm that it is publicly accessible. The verified link must then be inserted on the **cover page** of your Business Analyst Report. Failure to provide a working Tableau Public link on the cover page will result in your dashboards being considered non-submitted.

Part A – Tableau Story

General Requirements

Your Tableau Story must:

- Contain exactly three dashboards.
- Follow a stepwise narrative, each dashboard builds logically on the previous one.
- Apply a global ESG filter so that only firm-years with a non-null ESG Score (ESGS) are included in every visual. This is a hard requirement and is the first thing markers will check.
- Include interactivity: global filters, highlight actions, and at least one dashboard-to-dashboard action.
- Incorporate Tableau's time-series forecast on at least one chart (Dashboard 3).
- Include a parameter-driven What-If analysis (user-adjustable scenario).
- Use captions: each story point must include a 1–2 sentence caption written by you, summarising the key insight. Example caption style (do not copy):

“Average ESG scores improved steadily across Asia-Pacific after 2015, with developed-market firms (AU, NZ, SG, KR) outpacing emerging-market peers.”

Required Calculated Fields and Parameter

Create the following calculated fields in your Tableau workbook before building the dashboards. Each field is referenced by name in the dashboard specifications below.

Calculated field	Formula	Used on
Market Group	IF [CountryofExchange] IN ("Australia","New Zealand","Singapore","Korea; Republic (S. Korea)") THEN "Developed APAC" ELSE "Emerging APAC" END	Dashboard 1 Dashboard 2
Has Controversy	IF [CONT] < 100 THEN "Yes" ELSE "No" END	Dashboard 2
Above Threshold	IF [ESGS] >= [ESG High Threshold] THEN "High" ELSE "Low" END	Dashboard 3 (driven by ESG High Threshold parameter)
ROA	SUM([ebit]) / SUM([at])	Dashboard 1 (KPI), Dashboard 3
Tobin's Q	(SUM([MCAP])/1000000 + SUM([lt])) / SUM([at])	Dashboard 1 (KPI), Dashboard 3
MTB	(SUM([MCAP])/1000000) / SUM([ceq])	Dashboard 3

Currency note: MCAP is reported in USD, while Compustat balance-sheet items (*at*, *lt*, *ceq*, *sale*, *ebit*, *ebitda*) are reported in USD millions. Divide MCAP by 1,000,000 before combining with balance-sheet items in calculations such as Tobin's Q and MTB. Failing to do so will produce values in the trillions — markers will check.

Parameter: ESG High Threshold

Parameter	Type	Range	Default	Used on
ESG High Threshold	Integer	50 to 80, step 5	70	Dashboard 3 (drives Above Threshold field)

Dashboard 1 – Asia-Pacific Market Landscape & ESG Trends (2010–2024)

Purpose: Provide an overview of firms by country, industry and ESG performance across the nine Asia-Pacific markets and establish the regional baseline against which Dashboards 2 and 3 build.

Required Elements:

- **KPI table:** Avg ESG Score, Avg Environmental / Social / Governance Pillar Scores, Total MCAP (USD billions), Avg ROA, Avg Tobin's Q. Each KPI in a separate labelled cell, calculated across the full filtered panel (all firm-years, 2010–2024). Where outliers materially distort an average (e.g. ROA), show the median alongside or annotate the value. [Purpose - Snapshot of regional ESG and financial performance.](#)
- **Line chart:** Average ESG Score over time, one line per Country of Exchange. Multi-select country filter shown on the dashboard; default to all nine countries. Use 'Show Missing Values' on the year axis. [Purpose - Trajectory and convergence/divergence across markets.](#)
- **Filled map (choropleth):** Total MCAP by Country of Exchange, with log colour scale or values in USD billions so smaller markets remain visible. Tooltip: country name and total MCAP. [Purpose - Geographic concentration of market value.](#)
- **Horizontal bar chart:** Top 10 GICS Industries by total MCAP, sorted descending, coloured by average ESG Score. Tooltip: industry, MCAP (USD billions), avg ESG. [Purpose - Whether the largest sectors lead or lag on ESG.](#)
- **Stacked horizontal bar chart:** For each Country of Exchange (one bar per country), show the share of the country's total MCAP held by each industry. Use GICSIndustryGroupName (24 groups) rather than GICSIndustryName (53 industries) so that the segments remain readable. Apply a percent-of-total table calculation so each bar sums to 100%. Sort countries by total MCAP descending. [Purpose - Reveals industry-composition differences across markets that raw-total comparisons obscure.](#)

Dashboard 2 – ESG & Governance Drivers

Purpose: Explore governance and structural factors that explain variation in ESG performance.

Required Elements:

- **Stacked bar chart:** Average Environmental, Social, and Governance pillar scores by GICS Industry Group, sorted by overall ESG descending. Each bar segmented by

pillar (E / S / G). **Purpose:** Reveals which pillar drives overall ESG variation across industries — and which lags.

- **Scatter plot:** Average ESG Score (y-axis) vs Average Female on Board % (x-axis), one point per firm. Points coloured by industry, sized by Market Cap. Add a linear trend line and display R^2 . Confirm the x-axis range stays within 0–100; FOB is already a percentage and must not be re-aggregated as a sum. **Purpose:** Tests whether board gender diversity is associated with stronger ESG performance.
- **Heatmap:** Average ESG Score by Country (rows) $\times$ Year (columns). Use a sequential colour scale. Annotate cells with the score where space allows. **Purpose:** Surfaces which Asia-Pacific markets are catching up, stagnating, or where coverage is thinnest.
- **Governance comparison:** Distribution of ESG Scores for firms with CEO Duality (CEOD = 1) versus without (CEOD = 0), shown as side-by-side boxplots or violin plots. **Purpose:** Tests whether separation of CEO and Chair roles is associated with stronger ESG — a recurring debate in Asia-Pacific governance reform.
- **Controversy chart:** Count (or proportion) of firm-years with active controversies (CONT < 100) by GICS Industry Group, sorted descending. Caption must note explicitly that the LSEG Controversies score is on an inverse scale (100 = no controversy, lower = more severe). **Purpose:** Identifies industries where reputational and regulatory risk concentrates regardless of overall ESG score.

Dashboard 3 – ESG, Governance & Financial Performance

Purpose: Link ESG to firm financial outcomes, allow forecasting, and enable scenario testing. Move from drivers (Dashboard 2) to consequences and projections.

Required Elements:

- **Scatter plot:** ROA (y-axis) vs ESG Score (x-axis), with one point per firm-year (do not aggregate to industry). Points coloured by GICS Industry Group, sized by Market Cap. Filter ROA to the range -1 to $+1$ to remove extreme outliers; annotate the filter on the chart. Add a linear trend line and report R^2 . **Purpose:** Tests the central empirical question — is firm-level ESG performance associated with higher financial returns in Asia-Pacific?
- **Dual-axis line chart with forecast:** Average ESG Score and Average Tobin's Q over time (2010–2024) with Tableau's forecast extended to 2028, including 95% confidence intervals on both lines. Caption must quantify the uncertainty - e.g., “by 2028 the ESG forecast spans a range of approximately $\pm X$ points, indicating low

predictive confidence” - not merely note that uncertainty exists. **Purpose:** Examines whether ESG and valuation move together over time and surfaces the actual level of uncertainty in any forward-looking projection.

- **Bar comparison — Above vs Below ESG Threshold:** Average ROA and Average MTB for firms above the **ESG High Threshold** versus firms below it (using the **Above Threshold** calculated field — see parameter block below). Shown as side-by-side or grouped bars. **Purpose:** Provides a discrete, interpretable answer to the ESG-performance question for audiences less comfortable with scatter plots.
- **Country-level High-vs-Low ESG comparison:** For each of the nine countries, compare Average ROA and Average Tobin's Q for firms above the ESG High Threshold versus firms below it (using the same **Above Threshold** calculated field). Display as a grouped bar chart or small-multiples view. **Purpose:** Tests whether the ESG-performance relationship holds uniformly across the region or whether it depends on market maturity and regulatory environment.

What-If Parameter — ESG High Threshold

Create a parameter:

- **Name:** ESG High Threshold
- **Range:** 50 to 80
- **Step:** 5 (so users can select 50, 55, 60, 65, 70, 75, or 80)
- **Default:** 70

Create a calculated field driven by the parameter:

- **[Above Threshold = IF [ESGS] >= [ESG High Threshold] THEN "High" ELSE "Low" END]**

Apply **Above Threshold** to the last two charts (the bar comparison and the country-level High-vs-Low comparison). The parameter slider must be visible as a control on the dashboard so users can adjust it interactively.

Worked example: A firm with an ESG Score of 72 is classified as "High" when the threshold is set to 70 (because $72 \geq 70$) but becomes "Low" when the threshold is raised to 75 (because $72 < 75$). The firm's actual ESG performance hasn't changed — only the cutoff defining "High" has. Moving the slider re-classifies firms across the region and updates the last two charts accordingly.

Part B – Written Business Analyst Report (1,000 words ±10%)

Purpose

The Business Analyst Report complements your Tableau Story by transforming visual insights into a structured, evidence-based narrative. While the dashboards provide interactive analysis, the report demonstrates your ability to:

- Interpret complex data by extracting meaning from trends, patterns and anomalies across nine Asia-Pacific markets.
- Critically evaluate findings by addressing data quality limitations, forecasting uncertainty, and gaps in ESG and governance disclosure across markets.
- Integrate external evidence by drawing on academic literature, regulatory developments across the region (ASIC and AASB in Australia; NZ FMA and XRB in New Zealand; MAS in Singapore; SEBI in India; CSRC and HKEX in China; KSSB in Korea), aligned with global ISSB standards, and industry reports.
- Provide actionable recommendations for firms, investors and regulators.
- Communicate professionally in a concise business-style format, mirroring consulting or analyst deliverables.

To ensure consistency and academic rigour, your report must follow the required structure below:

1. Executive Summary (150–200 words)

- Concise overview of findings and key implications.
- Must stand alone — a manager should understand your main message without reading the full report.

2. Introduction (100–150 words)

- Purpose of the report and analysis.
- Why ESG and governance matter in the Asia-Pacific corporate context (developed vs emerging markets, mandatory ISSB-aligned disclosure regimes).
- Brief reference to macroeconomic and regulatory background.

3. Findings & Critical Analysis (500–550 words)

- Dashboard 1: Asia-Pacific market overview and ESG trends (key KPIs, industries, country and developed-vs-emerging differences, industry-mix variation across markets).

- Dashboard 2: Governance drivers of ESG (board gender diversity via FOB, CEO duality, controversies, and industry effects).
- Dashboard 3: Links between ESG and financial outcomes (ROA, Tobin's Q, MTB, including the forecast to 2028 and the ESG High Threshold sensitivity test).
- Critical reflection:
 - Reliability and limitations of LSEG/Refinitiv ESG measures (provider disagreement, backfilling, size bias).
 - Impact of missing values, outliers and uneven reporting (e.g. ESG scores are concentrated in larger firms; Bangladesh and NZ have small samples).
 - Forecasting uncertainty (width of 95% confidence intervals to 2028).
 - Assumptions and limitations of the ESG High Threshold parameter (definitional choices, sensitivity of findings to where “High ESG” is drawn, implications of small markets having few firms above higher thresholds).
- Integration of external sources (minimum 5–6 references from academic literature, regulator statements, or industry analysis)

4. Recommendations (150–200 words)

- **Firms:** Strengthen governance (board diversity, board independence, separation of CEO and Chair roles where appropriate), manage controversies proactively, improve disclosure aligned with ISSB / local standards.
- **Investors:** Incorporate ESG into valuation and risk analysis; account for cross-country differences in disclosure quality and rating-threshold conventions when comparing firms.
- **Policy/Regulators:** Link findings to regional supervisory frameworks — ASIC/AASB and NZ FMA/XRB in ANZ, MAS in Singapore, SEBI BRSR Core in India, CSRC and HKEX in China, KSSB in Korea — and to ISSB convergence; consider implications for cost of capital and investor confidence.

5. Conclusion (50–100 words)

- Reinforce the ESG–performance relationship in the Asia-Pacific context.
- Summarise long-term implications for corporate strategy, investors and regulators.

Important:

- All submissions must be made via Blackboard. Submissions by email will not be accepted.
- Ensure that your file or link is accessible and fully playable before the deadline.
- Late submissions will be managed in accordance with Curtin's Assessment and Student Progression Manual.

Marking criteria

This assessment will be evaluated using the marking criteria and rubric outlined below. The rubric ensures alignment with the intended learning outcomes and provides clear guidance on the standards expected. It sets out how marks will be allocated across the Tableau Story (70%) and the Business Analyst Report (30%), with detailed descriptors to distinguish between levels of performance

Marking Rubric: Part A – Tableau Story (70%)				
Criterion	Below Expectations	Meets Expectations	Exceeds Expectations	Proficient (Distinction/HD)
Dashboard Design & Presentation (20%)	Dashboards incomplete or missing required elements. Poorly formatted visuals. ESG filter not applied.	Some required charts included but formatting is basic and lacks professional finish. ESG filter applied inconsistently	All required charts included. ESG filter applied globally. Dashboards are well-structured and clear, with appropriate data-quality annotations on key metrics.	Dashboards are polished, professional and visually compelling. All required elements included and formatted consistently. KPIs and visuals provide immediate insight, with explicit data-quality annotations where outliers or coverage gaps are present.
Story Progression & Captions (15%)	Dashboards disconnected, no logical sequence. Captions missing or irrelevant.	Some progression visible but flow weak. Captions are included but descriptive only.	Dashboards build logically (overview → drivers → outcomes). Captions summarise main findings.	Seamless narrative flow across the three dashboards. Captions are concise and insightful, advancing a defensible analytical claim grounded in the visual rather than merely describing it. Captions connect observations to broader regulatory or market context where relevant, building a coherent story from overview through drivers to outcomes
Interactivity & Engagement (10%)	No filters, highlights, or dashboard actions. Dashboards static.	Basic interactivity (e.g., one filter). Limited user engagement.	Multiple interactive features (filters, highlights, dashboard actions) functional and relevant.	Highly interactive dashboards with global filters, highlights and dashboard actions that meaningfully enhance exploration. At least one cross-dashboard action (e.g., clicking a country in Dashboard 1 filters, Dashboards 2 and 3). Interactivity is intuitive and adds analytical value rather than being decorative
Forecasting (Time-Series) (10%)	No forecasting attempted, or incorrect use of Tableau's forecasting tool.	Basic forecast included but poorly integrated. No interpretation.	Forecast applied correctly (e.g., ESG vs Tobin's Q) and linked to analysis. Trends discussed.	Forecasting applied appropriately with quantitative interpretation of trends and 95% confidence intervals. Forecast extends insight to 2028 and is critically discussed.
Parameter-Driven What-If Analysis (15%)	No parameter created, or parameter non-functional.	Parameter created but applied superficially (e.g., slider exists but weakly linked to visuals).	ESG High Threshold parameter works correctly and modifies the bar comparison and country-level chart. Impact demonstrated.	Parameter is well-designed, integrated across the relevant charts on the dashboard, and used to simulate meaningful scenarios. Impacts on the High/Low classification and the resulting averages are visibly demonstrated, including for small markets where few firms qualify at higher thresholds.
Marking Rubric: Part B – Business Analyst Report (30%)				
Criterion	Below Expectations	Meets Expectations	Exceeds Expectations	Proficient (Distinction/HD)
Findings & Critical Analysis (Report) (15%)	Report descriptive only, summarises dashboards without interpretation. No external sources.	Some interpretation of dashboards. Limited reflection on data quality. One external source cited.	Clear analysis linking dashboards to ESG, governance, and financial outcomes. Reflection on ESG measures and data limitations. Supported by 2–3 relevant sources.	Sophisticated analysis integrating dashboards with academic literature, regulator reports (ASIC/AASB/NZ FMA/MAS/SEBI/CSRC/ISSB/ HKEX and KSSB) and industry insights. Critically

				evaluates ESG measures, forecasting uncertainty and What-If assumptions.
Recommendations (Report) (10%)	No recommendations, or vague and generic statements.	Basic recommendations provided, weakly linked to findings.	Clear recommendations for firms, investors, and regulators, linked to findings.	Insightful, actionable recommendations with explicit links to Asia-Pacific regulatory context (ASIC/AASB/NZ FMA/MAS/SEBI/CSRC/HKEX/KSSB/ISSB).
Professional Communication (Report) (5%)	Report poorly structured, weak writing, missing or inaccurate references.	Report adequate but with issues in clarity, flow, or referencing.	Report well-structured, clearly written, and properly referenced. Presentation professional.	Report polished and professional. Writing concise and academic. Clear logical flow, correct referencing, and suitable for senior executive or regulator audience.

Appendix A – Data Preparation in Tableau

The dataset (Asia-Pacific listed companies, 2001–2024) reflects real-world corporate disclosures and, as such, contains missing values, outliers and uneven coverage across countries and time. This is typical of corporate reporting environments and is intentionally preserved to ensure the assessment reflects authentic business analysis practice.

You are not expected to undertake detailed external data cleaning. Data preparation was the focus of Assessment 2. For Assessment 3, you must manage data quality issues within Tableau so that your dashboards are professional, interpretable and analytically robust.

Minimum Data Preparation Requirements in Tableau

1. Apply the ESG filter (mandatory first step)

- Add a global data-source filter: ESGS IS NOT NULL. This reduces the panel from ~237,506 firm-years to ~17,368 firm-years — a manageable size for Tableau Public and ensures every chart is built on the same ESG-disclosing sub-sample.

To add, a global data-source filter: top menu Data → [your data source name] → Edit Data Source Filters → Add → EsGs → All values → Next → Special tab → Non-null values → OK. This excludes firm-years with missing ESG scores from every dashboard automatically.

- Restrict the date axis to 2010–2024 (ESG coverage in this dataset effectively starts around 2010 for emerging Asia-Pacific markets).

2. Show Missing Values in Time Series

- Right-click the date axis and select Show Missing Values to maintain a continuous 2010–2024 timeline, particularly for smaller markets such as Bangladesh and New Zealand.

3. Filter Extreme Outliers

- Remove clearly unrealistic values (e.g. negative total assets, market cap below USD 1m, ROA outside $[-1, 1]$).
- Reference any anomalies in your report as part of your reflection on data quality.

4. Handle Nulls Transparently

- Replace categorical nulls (e.g. CEOD, GICSIndustryName) with “Not Reported.”
- For numeric nulls (e.g. EPS, SPS, GPS), leave as missing but discuss their impact (governance scores have higher coverage than environmental scores in many markets).

5. Use Simple Calculated Fields

- Create the calculated fields and parameter listed in Part A before building any dashboard.
- Where nulls prevent calculation, return “Not Available” rather than excluding firms entirely.

6. Currency note

- MCAP in this dataset is reported in USD, while Compustat balance-sheet items (at, lt, ceq, sale, ebit, ebitda) are reported in USD millions. Convert MCAP to USD millions before combining with balance-sheet items in calculations such as Tobin’s Q.

7. Controversy score scale

- The LSEG/Refinitiv Controversies score (CONT) uses an inverse scale: 100 = no controversy, lower values indicate a controversy occurred

Note: You are not assessed on the perfection of data cleaning, but on how effectively you manage and reflect on data quality issues in Tableau.

Appendix B: Suggested Sources for Support

Australian Securities & Investments Commission (ASIC):

- **ASIC Regulatory Guide RG 280: Sustainability Reporting:** ASIC has published final guidance for entities required to prepare sustainability reports containing climate-related financial disclosures [Link: [Here](#)]
- **ASIC Mandates for Climate Reporting from 1 January 2025:** [Click [Here](#)]

Australian Accounting Standards Board (AASB):

- AASB & AUASB Climate-Related and Sustainability Disclosure Standards (aligned with ISSB standards, 2023–24).

Reserve Bank of Australia (RBA):

- Statements on monetary policy and Financial Stability Reviews, which highlight ESG and climate risks in capital allocation.

Financial Markets Authority (FMA – New Zealand):

- Climate-Related Disclosures Regime Guidance (mandatory reporting from 2023 under NZ’s Financial Sector (Climate-Related Disclosures and Other Matters) Amendment Act 2021).

Other Asia-Pacific Markets

- **Singapore — MAS / SGX RegCo:** SGX Sustainability Reporting Guide and climate-related disclosure rules; MAS Guidelines on Environmental Risk Management. [Click [Here](#)] [Click Here]
- **India — SEBI:** Business Responsibility and Sustainability Report (BRSR) and BRSR Core (mandatory for the top 1,000 listed entities). [Click [Here](#)]
- **China — CSRC and HKEX:** Sustainability Disclosure Guidelines for listed companies (Shanghai/Shenzhen/Beijing); HKEX Climate-related disclosure requirements aligned with ISSB. [Click [Here](#)]
- **Korea — KSSB:** Korea Sustainability Standards Board exposure drafts on sustainability and climate disclosures. [Click [Here](#)]
- **Malaysia — Bursa Malaysia:** Sustainability Reporting Guide and Main/ACE Market Listing Requirements on sustainability disclosure. [Click [Here](#)]
- **Indonesia — OJK:** POJK No. 51/2017 on Sustainable Finance and Sustainability Reporting.
- **Bangladesh — BSEC:** Corporate Governance Code and emerging ESG disclosure expectations.

Global / Cross-Border Frameworks

- **IFRS Foundation — ISSB Standards** (IFRS S1 General Sustainability and IFRS S2 Climate).
- **TCFD** recommendations (now consolidated into IFRS S2).
- **GRI Standards** for sustainability reporting.

Industry & Practitioner Reports

- **MSCI ESG Research** (for global ESG rating methodology and sector comparisons).
- **Refinitiv / LSEG ESG database documentation** (for definitions of ESG, EPS, SPS, GPS, and controversies).
- **KPMG / Deloitte / PwC / EY Asia-Pacific sustainability and governance surveys**

Academic Literature (examples to guide citation)

- Aouadi, A., & Marsat, S. (2018). Do ESG controversies matter for firm value? Evidence from international data. *Journal of business ethics*, 151(4), 1027-1047.
- Chhaochharia, V., & Grinstein, Y. (2007). Corporate governance and firm value: The impact of the 2002 governance rules. *the Journal of Finance*, 62(4), 1789-1825.

- Jain, T., & Zaman, R. (2020). When boards matter: The case of corporate social irresponsibility. *British Journal of Management*, 31(2), 365-386.
- Jain, T., Zaman, R., & Harjoto, M. (2024). Behavioral agency model and corporate social irresponsibility: Uncovering the implication of fairness in CEO compensation. *Journal of Management*, 50(7), 2715-2754.
- Jo, H., & Harjoto, M. A. (2011). Corporate governance and firm value: The impact of corporate social responsibility. *Journal of business ethics*, 103(3), 351-383.
- Zaman, R., Atawnah, N., Banigidadmath, D., Nadeem, M., & Liu, J. (2025). Do companies' green credentials enhance trade credit provisions? Global evidence. *Journal of International Financial Markets, Institutions and Money*, 103, 102204.
- Zaman, R., Fatima, U., Farooq, M. B., & Kazemian, S. (2025). Co-opted directors and corporate climate risk disclosure. *Meditari Accountancy Research*, 33(7), 118-156.
- Zaman, R., Nadeem, M., & Carvajal, M. (2021). Corporate governance and corporate social responsibility synergies: evidence from New Zealand. *Meditari Accountancy Research*, 29(1), 135-160.